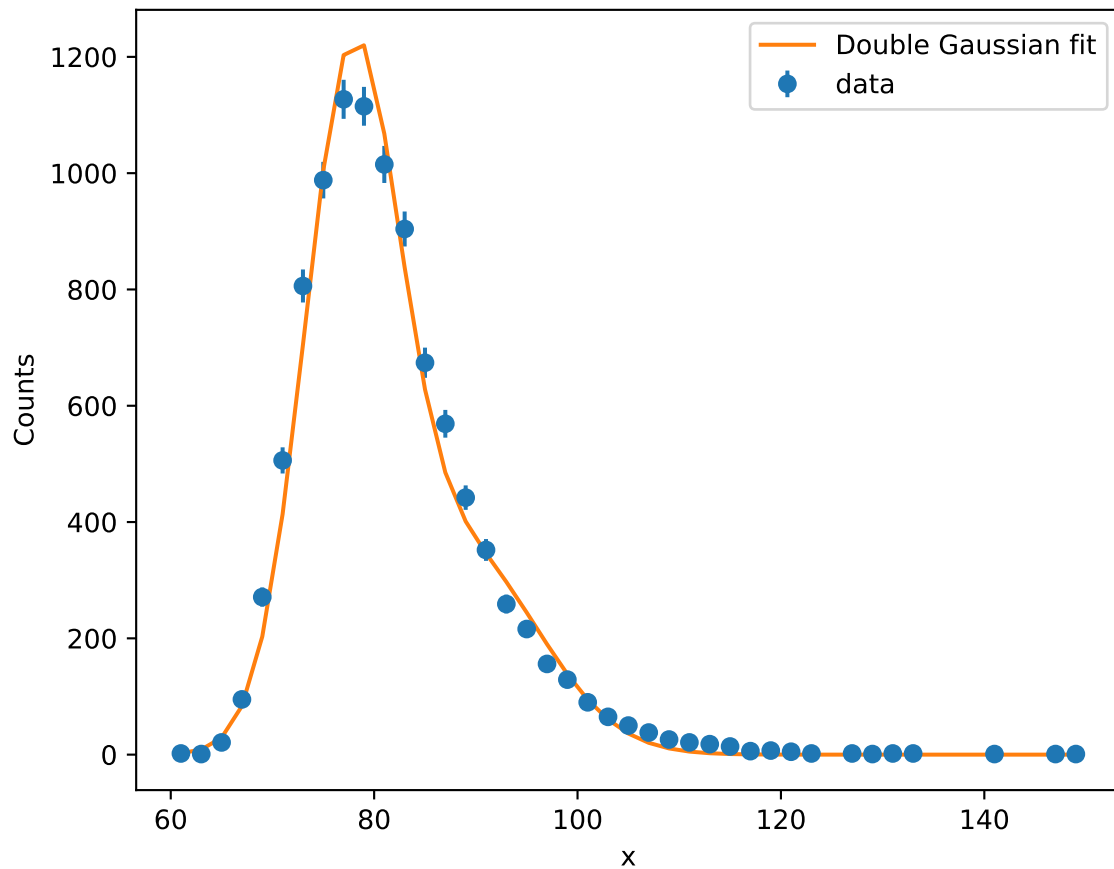
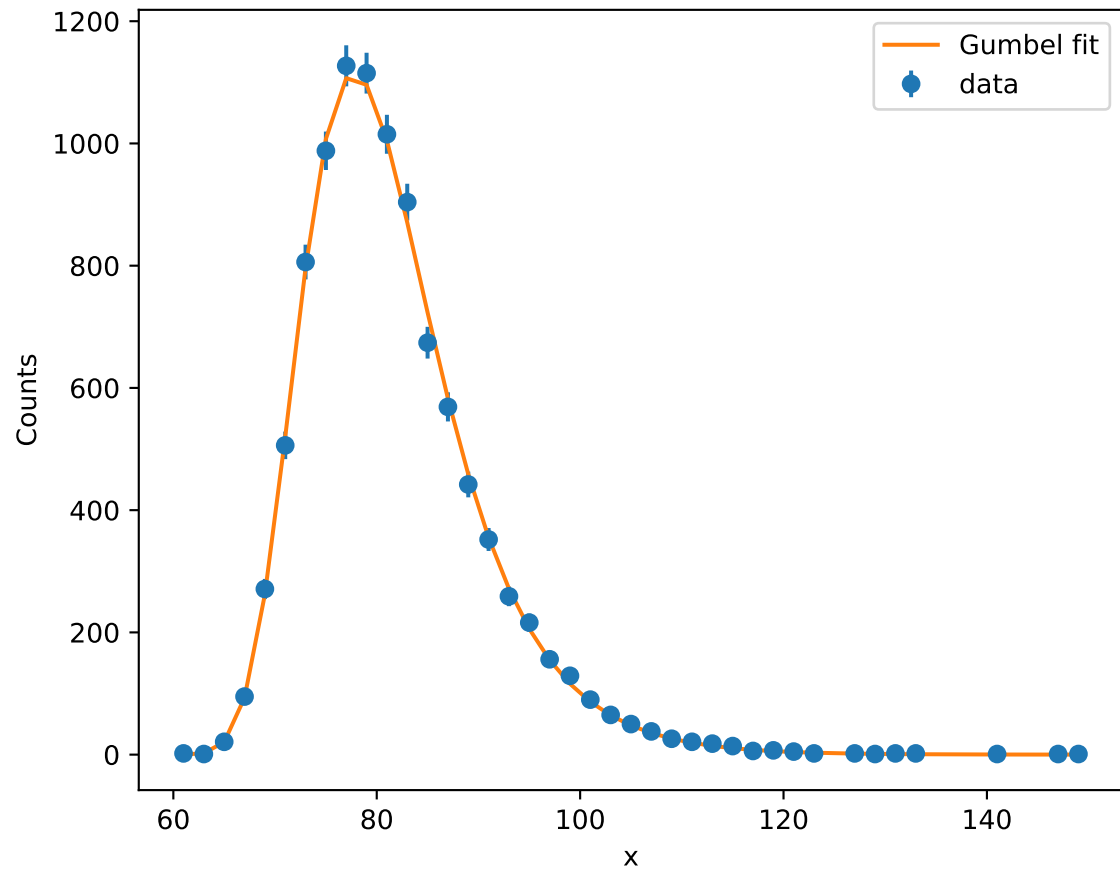


Double Gaussian Fit



Gumbel Fit



Fit Results Summary

Chi-square fits:

Double Gaussian:

chi2 = 258.05, dof = 33, p-value = 0.000

Gumbel:

chi2 = 21.51, dof = 36, p-value = 0.973

Comments:

- The Double Gaussian fit has an extremely low p-value (~ 0), meaning it is a poor fit to the data.
- The Gumbel fit has a very high p-value (~ 0.97), indicating an excellent agreement with the data.
- Based on chi2 and p-values, the Gumbel model is strongly preferred over the Double Gaussian model.

NLL-based fits (Poisson likelihood):

Double Gaussian: NLL = 22.89

Gumbel: NLL = 56.18

Delta NLL (Gumbel - Gaussian) = 33.29

Comments:

- Since $\Delta \text{NLL} = \text{NLL}(\text{Gumbel}) - \text{NLL}(\text{Gaussian}) > 0$, the Gaussian fit has slightly lower NLL.
- However, chi2 and p-values strongly favor the Gumbel fit, so overall the Gumbel model is preferred.